EARTHING DESIGN CRITERIA & IS/IEC CONFORMANCE:

1. Chemical earthing pits with 250-micron copper-bonded steel earth electrodes (Dia 17.2mm x 3000mm length).
2. Earth grid conductor: 50x6 mm electrolytic copper tape buried at 600mm depth in ring configuration around the substation.
3. Earth resistance value strictly maintained < 1.0 Ohm for electrical body earthing and < 0.5 Ohm for clean instrumentation.
4. Dedicated isolated clean earth pits provided for Data Center UPS, Fire Alarm Panel, and Audio/Broadcast systems.
5. Early Streamer Emission (ESE) lightning protection air terminal installed at tower peak with 107m protection radius (Level 1 per IS/IEC 62305).

Earth Pit Tag	Service Connection	Electrode Type	Strip Size	Max Resistance
EP-01 to 04	11kV Transformer Neutral	Copper Bonded 3m Rod	50x6 mm Copper Tape	< 0.8 Ohm
EP-05 to 08	Main LT Panel Body Earth	Copper Bonded 3m Rod	50x6 mm Copper Tape	< 1.0 Ohm
EP-09 to 12	DG Set 1 & 2 Neutral	Copper Bonded 3m Rod	50x6 mm Copper Tape	< 0.8 Ohm
EP-13 to 16	Clean ELV / BMS / Server	Pure Copper Plate Earth	50x6 mm Copper Tape	< 0.5 Ohm
EP-17 to 20	Lightning Protection Pit	Chemical Compound Rod	50x6 mm Copper Tape	< 1.0 Ohm

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: SUBSTATION EARTHING GRID & LIGHTNING PROTECTI		
DRAWING NO: SL-288-EL0-050-EARTHING-SI	REVISION SCHEMA	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	